

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 1

CANDIDATE
NAME

CLASS

TUTOR'S
NAME

CHEMISTRY

Paper 2

8872/02

19 September 2013

2 hours

Candidates answer Section A on the Question Paper.

Additional Materials: Answer Paper
 Data Booklet
 Graph Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** the questions.

Section B

Answer **two** questions on separate answer paper.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
B5	
B6	
B7	
Total	

This document consists of **15** printed pages.

[Turn over

Section A

Answer **all** the questions in this section in the spaces provided.

1 (a) Silicon and sulfur are among the most abundant period 3 elements on Earth.

(i) Describe the variations in melting points of the elements silicon and sulfur. Explain the variations in terms of their structures and bonding.

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(ii) Describe the action of water on the oxides of silicon and sulfur, write equations for any reactions that occur, and suggest the pH of each solution formed.

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[5]

(b) Sulfur hexafluoride (SF_6) is a colourless and odourless gas that is poorly soluble in water. It is hypothesised that sulfur pentafluoride ion (SF_5^-) is very soluble in water.

(i) Draw a dot-and-cross diagram to show the bonding in a sulfur pentafluoride ion.

(ii) State and explain in detail how the shape of sulfur pentafluoride ion arises.

shape

explanation

(iii) Explain with the aid of a diagram why sulfur pentafluoride ion is very soluble in water.

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[5]

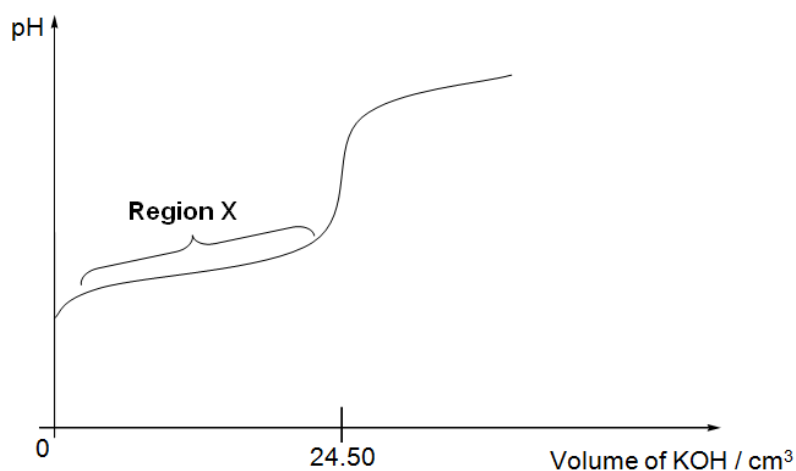
[Total: 10]

- 2 (a) Given the following information, explain why the different compounds have different pK_a values at 298 K.

Compound	pK_a
Benzoic acid	4.19
2-methylbutanoic acid	4.80
Butan-2-ol	17.6

[4]

- (b) 25.0 cm³ of 2-methylbutanoic acid was titrated against 0.15 mol dm⁻³ of KOH and the following titration curve was obtained.



- (i) Explain what is meant by the term acid dissociation constant, K_a , of 2-methylbutanoic acid.

- (ii) Calculate the concentration of 2-methylbutanoic acid used in this experiment.

- (iii) Using relevant values from (a) and b (ii), calculate the initial pH of 2-methylbutanoic acid acid.

- (iv) Suggest and explain a suitable indicator for the above titration.

indicator

explanation

- (v) With the aid of an equation, explain why the pH of the titration curve at region **X** remains fairly constant.

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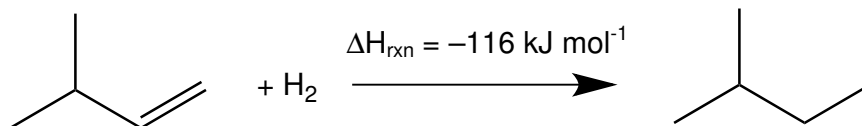
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[6]

[Total: 10]

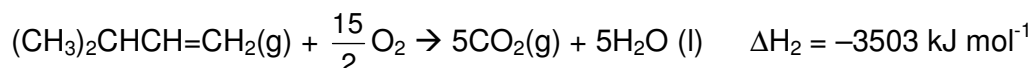
- 3 2-methylbutane is a component of natural gas. It is used as a fuel and also as an active ingredient in toothpaste for sensitive teeth.

2-methylbutane ($M_r = 72.0$) can be synthesised from 3-methylbut-1-ene via a reaction with hydrogen under suitable conditions. The enthalpy change of this reaction is -116 kJ mol^{-1} .



- (a) (i) Suggest the conditions for this reaction.

- (ii) Calculate the enthalpy change of combustion of 2-methylbutane given the following enthalpy changes:

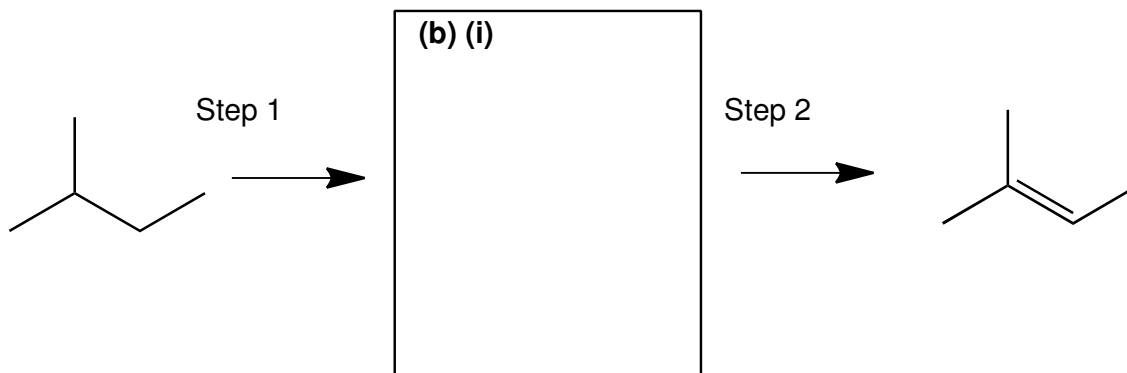


- (iii) Calculate the minimum mass of 2-methylbutane required to heat 1000 g of water from 25°C to boiling point.

The specific heat capacity of water is $4.2 \text{ J g}^{-1} \text{ K}^{-1}$.

[5]

- (b) Refer to the reaction scheme of the synthesis of 2-methylbut-2-ene from 2-methylbutane below.



- (i) Draw the structure of a possible intermediate for the reaction in the space above.

- (ii) State the type of reaction occurring in steps 1 and 2.

- (iii) Give reagents and conditions for steps 1 and 2.

Step 1

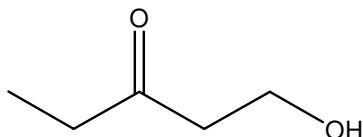
Step 2

- (iv) Will the yield of 2-methylbut-2-ene be high? Explain your answer.

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 [5]

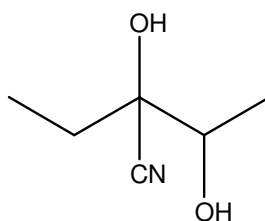
[Total: 10]

- 4 Pheromones are compounds an animal produces which can influence a change in the behavior of another animal or insect of the same species. Compound **A** is an isomer of an insect pheromone with a distinct peanut and hay-like odour characteristic.



Compound **A**

- (a) In not more than 3 steps, suggest a synthetic route for the formation of **B** from **A**. State clearly the reagents and conditions for each reaction step.



Compound **B**

[5]

- (b) Compound **B** can undergo a single reaction to form an amine. Draw the displayed formula of this amine and state the type of reaction and the reagents and conditions used in this reaction.

Type of reaction

Reagents and conditions

.....
[3]

- (c) Suggest the reagents and conditions for a chemical test that could be used to distinguish between compounds **A** and **B**.

Reagents and conditions

Observations

.....
[2]

[Total: 10]

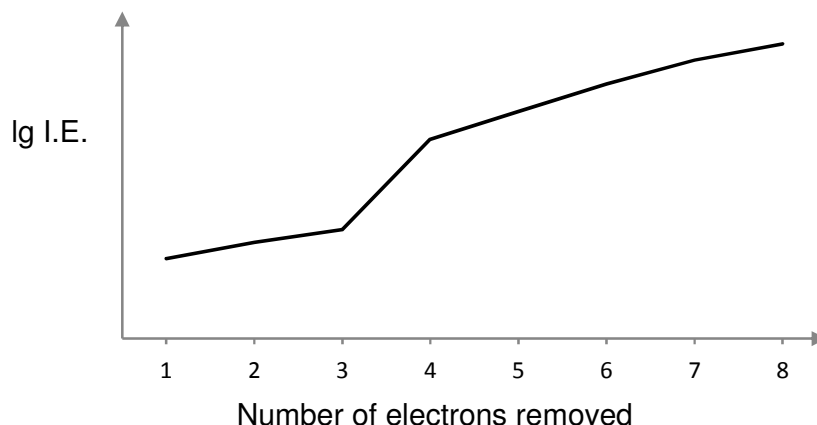
Section B

Answer **two** questions from this section on separate answer paper.

- 5 (a) Sketch a graph to show how ionic radius changes across period 3 elements and explain the shape of your graph. [3]

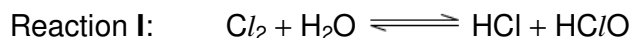
- (b) (i) With the aid of an equation, define the 2nd ionisation energy of gallium.

The graph below shows the 1st 8 ionisation energies of element **D**.



- (ii) Prove with reasoning that Element **D** belongs to the same group as gallium in the Periodic Table. Write the electronic configuration of the outer shell of element **D**.
- (iii) Gallium(III) oxide is used as a precursor in the manufacturing of semiconductor materials. Draw the dot-and-cross diagram of gallium(III) oxide. [6]
- (c) Period 3 oxides can be formed by combustion of the elements in oxygen. Describe the formation of sodium oxide and phosphorus pentoxide and write down the relevant equations of the reactions. [3]

- (d) The addition of Cl_2 to water forms hydrochloric acid and hypochlorous acid according to the following equation.



Hypochlorous acid hydrolyses in water to form a hypochlorite ion, OCl^- . The hypochlorite ion formed is the main component in bleach and disinfectant agents.



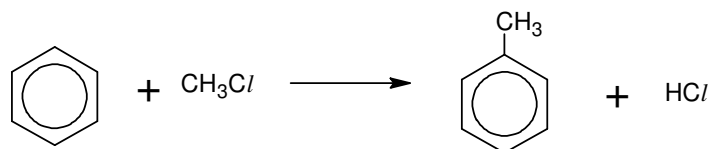
- (i) Explain why the formation of the stable hypochlorite ion is facilitated by dissolving chlorine gas in basic solutions, such as sodium hydroxide.
- (ii) 1 mole of Cl_2 was dissolved in 1 dm^3 of water with a dissociation coefficient of 0.75 in reaction I. Calculate the value of K_c , giving its units.
- (iii) There are two unlabelled solutions **X** and **Y**, each containing the same concentration of HCl (aq) and HOC l (aq). To identify the solutions a student mixed equal volumes of $NaOH$ (aq) with each solution. The following results are obtained:

Reaction	$\Delta T / ^\circ C$
$NaOH + X$	4.8
$NaOH + Y$	5.6

Deduce the acid present in solutions **X** and **Y** and explain your answer. [8]

[Total: 20]

- 6 (a) Methylbenzene can become 2-chlorobenzoic acid and 3-chlorobenzoic acid via different reaction synthesis. Give the reagents and conditions required for the synthesis of 2-chlorobenzoic acid, write the chemical equations and name the type of reactions that occur. State and explain how 3-chlorobenzoic acid may be synthesised instead of 2-chlorobenzoic acid. [6]
- (b) Explain why 2-chlorobenzoic acid is a stronger acid than benzoic acid. [2]
- (c) The equation for the formation of methylbenzene from benzene and chloromethane is shown.



A student performed an experiment to determine the order of reaction with respect to chloromethane. The initial concentration of chloromethane was $0.0100 \text{ mol dm}^{-3}$ and benzene was used in large excess. The concentration of chloromethane at various times in the reaction mixture was found as follows:

Time / s	[chloromethane] / mol dm^{-3}
0	0.0100
40	0.0079
80	0.0062
120	0.0049
160	0.0038
200	0.0030

- (i) Explain what is meant by *order of reaction*.
- (ii) Plot the above data on suitable axes and use your graph to determine the order of reaction with respect to chloromethane.
- (iii) The initial concentration of chloromethane was $0.0100 \text{ mol dm}^{-3}$ and benzene was used in large excess. Suggest why benzene was used in large excess in this experiment?
- (iv) Explain with the aid of a suitable diagram, how the presence of a catalyst can cause an increase in the rate of a chemical reaction. [12]

[Total: 20]

7 Maleic acid is one of two stereoisomers of butenedioic acid, $\text{HOOC}(\text{CH})_2\text{COOH}$. Maleic acid is the *cis*-isomer whereas fumaric acid is the *trans*-isomer. This question examines the chemistry of maleic acid and its derivatives.

- (a) (i) Draw the full structural formula of maleic acid.
- (ii) The two stereoisomers have different melting points as intramolecular hydrogen bonding is possible only in one isomer. Identify this isomer and hence, explain whether it has a higher or lower melting point, compared to the other isomer. [4]
- (b) Maleic acid is not an approved additive as long term consumption of high levels of maleic acid can cause kidney damage. In May 2013, the Agri-food & Veterinary Authority of Singapore recalled several food products imported from Taiwan that were found to contain maleic acid. Most of the affected products were tapioca starch balls used in the preparation of bubble tea beverages.

Some physical and chemical properties of maleic acid are given below.

Molecular formula: $\text{C}_4\text{H}_4\text{O}_4$

Physical state and appearance: Powdered solid

Colour: White.

M_r : 116.08

Density: 1.59 g cm^{-3}

Acid/Base properties: Dibasic

Water/Oil Distribution Coefficient: The product is more soluble in water;

$$\lg \frac{[\text{in oil}]}{[\text{in water}]} = -0.5$$

In a volumetric analysis to determine percentage by mass of maleic acid present in tapioca starch balls, a 5.0 g sample was first powdered and dissolved in 100.0 cm^3 of an oil-based solvent. Most of the maleic acid was extracted from the oil-based solvent into an aqueous medium with 100.0 cm^3 distilled water. 25.0 cm^3 of the aqueous solution required 7.10 cm^3 of 0.05 mol dm^{-3} aqueous sodium hydroxide for complete neutralisation.

- (i) Using the data provided, determine the concentration of maleic acid present in 100.0 cm^3 of aqueous solution. You may assume the only acid present in tapioca starch balls is maleic acid.
- (ii) When a solute is added to two solvents, A and B, which do not mix, some of the solute dissolves in each of the solvents and an equilibrium is set up between the two solvents. It has been shown that for dilute solutions, the ratio of the two concentrations in each of the solvents at equilibrium is a constant known as the **Dissociation Coefficient**.

Using your answer in (i) and the data provided, determine the concentration of maleic acid remaining in the oil-based solvent after extraction.

- (iii) Hence, using your answer in (i) and (ii) and data provided, determine the percentage by mass of maleic acid in the sample of tapioca starch balls.

[7]

- (c) (i) Maleic acid can be reduced to a compound **S** with molecular formula $\text{C}_4\text{H}_8\text{O}_2$. State the reagents required for this conversion and draw the full structural formula of **S**.
- (ii) Compound **S** has other structural isomers **T** and **U** all having the molecular formula $\text{C}_4\text{H}_8\text{O}_2$.

1 mole of **T** gives 1 mole of yellow precipitate with aqueous alkaline iodine. **T** also decolourises aqueous bromine and exhibits geometric isomerism.

1 mole of **U** also gives 1 moles of yellow precipitate with aqueous alkaline iodine. However **U** does not decolourise aqueous bromine.

Deduce the structures of compounds **T** and **U**. Explain your reasoning clearly.

[9]

[Total: 20]